Sliding Window - Maximum Sum of Size K

Topic: CompetitiveProgramming File: c4.pdf

PROBLEM

Find the maximum sum among all contiguous subarrays of exactly k elements.

APPROACH

Sum the first k values. For each next position, subtract the element leaving the window and add the new element entering it.

EXAMPLE

A = [2,1,5,1,3,2], k=3. Window sums are 8, 7, 9, and 6; the maximum is 9 from [5,1,3].

COMPLEXITY

Time: O(n), because each element enters and leaves at most once. Extra space: O(1).

EDGE CASES

Validate 1 <= k <= n. If array values may be negative, initialize the best value from the first valid window, not zero.